

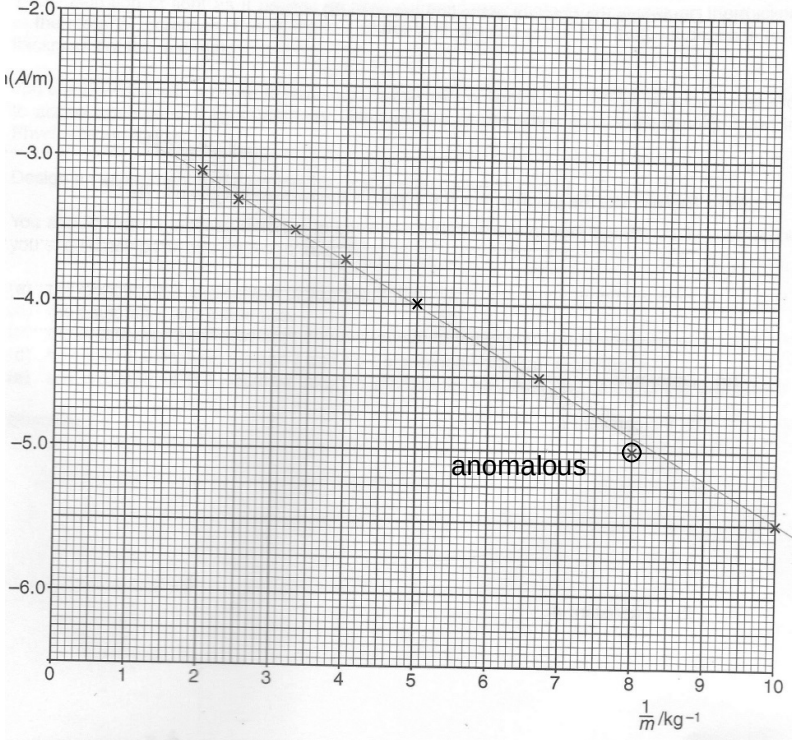
can be positive or negative depending on the polarity of the charge.

6(a)

The rate of change of A decreases with t , since the gradient decreases with t .

6(b)(i)

m/kg	$\frac{1}{m}/\text{kg}^{-1}$	$A/10^{-2} \text{ m}$	$\ln (A/m)$
0.200	5.00	1.8	-4.0

6(b)(ii)	
6(b)(iii)	$\begin{aligned}\text{Gradient} &= \frac{-5.400 - (-3.150)}{9.70 - 2.10} \\ &= \frac{-2.250}{7.60} \\ &= -0.30\end{aligned}$
6(b)(iv)	From $A = A_0 e^{-bt/2m}$, we get $\ln A = \ln A_0 + \left(-\frac{bt}{2}\right)\left(\frac{1}{m}\right)$ Fig.6.4 shows $\ln A$ varies linearly with $\frac{1}{m}$. The negative gradient of -0.30 is equal to $-\frac{bt}{2}$ and the vertical intercept is $\ln A_0$. Hence the graph supports the equation.
6(b)(v) 1.	$\begin{aligned}-\frac{bt}{2} &= \frac{-2.250}{7.60} \\ -\frac{b(5.0)}{2} &= \frac{-2.250}{7.60} \\ b &= 0.12\end{aligned}$
6(b)(v) 2.	$\begin{aligned}\ln A &= \ln A_0 + \left(-\frac{bt}{2}\right)\left(\frac{1}{m}\right) \\ -3.150 &= \ln A_0 + \left[\frac{-2.250}{7.60}\right] (2.10) \\ A_0 &= 8.0 \times 10^{-2} \text{ m}\end{aligned}$

6(c)	$A = A_0 e^{-bt/2m}$ $\frac{1}{2}A_0 = A_0 e^{-(0.12)t / 2(0.500)}$ $t = 5.8 \text{ s}$
6(d)	$T_{1/2}$ decreases linearly with m. Energy of the oscillation system is proportional to A^2. From $A = A_0 e^{-bt/2m}$ we get $\frac{1}{2}A_0^2 = A_0^2 e^{-bT_{1/2}/m}$ and $\ln\left(\frac{1}{2}\right) = -bT_{1/2}\left(\frac{1}{m}\right)$ Hence $T_{1/2} \propto m$
7	Aim: To determine the relationship between I and t